04-Model-Cards

1 Model Cards for Model Reporting

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Status: Emerging Best Practice

1.1 Table of Contents

1.2 Overview

Model Cards are short documents accompanying trained machine learning models that provide benchmarked evaluation in a variety of conditions, such as across different cultural, demographic, or phenotypic groups (e.g., race, geographic location, sex, age) and intersectional groups that are relevant to the intended application domains.

1.2.1 Key Features

- **Transparency**: Clear documentation of model capabilities and limitations
- **Accountability**: Track model lineage and responsible parties
- **Ethical AI**: Document bias testing and fairness considerations
- **Reproducibility**: Include training details and evaluation metrics
- **Governance**: Support regulatory compliance and auditing

1.2.2 Primary Use Cases

1. **Model Documentation**: Comprehensive model metadata
2. **Bias Assessment**: Document performance across demographic groups
3. **Compliance**: Meet regulatory requirements (EU AI Act, etc.)
4. **Model Comparison**: Standardized comparison across models
5. **Risk Management**: Identify and communicate model limitations

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1.3 Authoritative References

1.3.1 Original Paper

- **Title**: "Model Cards for Model Reporting"
- **Authors**: Margaret Mitchell, Simone Wu, et al. (Google)
- **Year**: 2019
- **Link**: https://arxiv.org/abs/1810.03993

1.3.2 Implementation Guides

- **Model Card Toolkit (Google)**: https://github.com/tensorflow/model-card-toolkit
- **Hugging Face Model Cards**: https://huggingface.co/docs/hub/model-cards
- **Microsoft Model Card**: https://www.microsoft.com/en-us/ai/responsible-ai-resources

1.3.3 Related Standards

- **EU AI Act**: Requires model documentation
- **IEEE 7000-2021**: Systems engineering for ethical concerns
- **ISO/IEC 23894**: AI Risk Management

1.4 Format Structure

Model Cards can be represented in **Markdown**, **JSON**, or **YAML**. Most implementations use Markdown for human readability.

1.4.1 Core Sections

```
graph TD
    A[Model Card] --> B[Model Details]
    A --> C[Intended Use]
    A --> D[Factors]
    A --> E[Metrics]
    A --> F[Evaluation Data]
    A --> G[Training Data]
    A --> H[Quantitative Analysis]
    A --> I[Ethical Considerations]
    A --> J[Caveats & Recommendations]
```

1.4.2 1. Model Details

Role: Basic model information and provenance.

Key Fields: - Model name and version - Model type/architecture - Organization/person developed - Model date - Model license - Contact information - Citation details

Impact: Establishes model identity and ownership.

1.4.3 2. Intended Use

Role: Defines appropriate use cases and users.

Key Fields: - Primary intended uses - Primary intended users - Out-of-scope uses - Use case examples

Impact: Prevents misuse and sets expectations.

1.4.4 3. Factors

Role: Relevant demographic/phenotypic factors.

Key Fields: - Groups (age, gender, geography, etc.) - Instrumentation (sensors, devices) - Environment (lighting, weather, etc.)

Impact: Identifies performance variation factors.

1.4.5 4. Metrics

Role: Evaluation metrics used.

Key Fields: - Model performance measures - Decision thresholds - Variation approaches (confidence intervals)

Impact: Quantifies model quality.

1.4.6 5. Evaluation Data

Role: Test set information.

Key Fields: - Datasets used - Motivation for dataset choice - Preprocessing steps

Impact: Enables result reproduction.

1.4.7 6. Training Data

Role: Training set information.

Key Fields: - Datasets used - Data collection methods - Preprocessing steps - Data splits

Impact: Documents model foundations.

1.4.8 7. Quantitative Analysis

Role: Performance across factors.

Key Fields: - Disaggregated evaluation - Intersectional analysis - Performance plots/tables

Impact: Reveals bias and limitations.

1.4.9 8. Ethical Considerations

Role: Ethical implications and risks.

Key Fields: - Data privacy considerations - Bias and fairness analysis - Potential harms - Mitigation strategies

Impact: Addresses responsible AI concerns.

1.4.10 9. Caveats and Recommendations

Role: Limitations and usage guidance.

Key Fields: - Known limitations - Edge cases - Recommendations for use - Monitoring guidance

Impact: Informs safe deployment.

1.5 Procedural Use-Cases

1.5.1 Use-Case 1: Creating a Model Card for Image Classification

Goal: Document a trained image classifier with bias analysis.

Workflow:

```
graph LR
    A[Train Model] --> B[Evaluate Performance]
    B --> C[Disaggregated Analysis]
    C --> D[Document Findings]
    D --> E[Create Model Card]
    E --> F[Publish with Model]
```

Step-by-Step Procedure:

1. **Collect Model Information**:

```
model_info = {
    "name": "ResNet-50 ImageNet Classifier",
    "version": "1.0.0",
    "architecture": "ResNet-50",
    "framework": "PyTorch 2.0",
    "developed_by": "AI Research Lab",
    "date": "2025-12-04",
    "license": "Apache 2.0"
}
```

2. **Document Training**:

```python
training_info = {
    "dataset": "ImageNet-1K",
    "samples": 1281167,
    "preprocessing": [
        "Resize to 256x256",
        "Random crop 224x224",
        "Horizontal flip (p=0.5)",
        "Normalization (ImageNet stats)"
    ],
    "hyperparameters": {
        "batch_size": 256,
        "learning_rate": 0.1,
        "epochs": 90,
        "optimizer": "SGD with momentum"
    }
}
```

3. **Perform Disaggregated Evaluation**:

```python
# Evaluate across different lighting conditions
evaluation_results = {
    "overall": {"accuracy": 0.762, "top5_accuracy": 0.931},
    "by_lighting": {
        "bright": {"accuracy": 0.812, "top5_accuracy": 0.951},
        "normal": {"accuracy": 0.762, "top5_accuracy": 0.931},
        "dark": {"accuracy": 0.631, "top5_accuracy": 0.847}
    },
    "by_object_size": {
        "large": {"accuracy": 0.854, "top5_accuracy": 0.967},
        "medium": {"accuracy": 0.762, "top5_accuracy": 0.931},
        "small": {"accuracy": 0.591, "top5_accuracy": 0.821}
    }
}
```

4. **Generate Model Card**:

```markdown
# Model Card: ResNet-50 ImageNet Classifier v1.0.0

## Model Details

- **Developed by**: AI Research Lab
- **Model date**: December 2025
- **Model type**: Convolutional Neural Network (ResNet-50)
- **Model version**: 1.0.0
- **License**: Apache 2.0
- **Framework**: PyTorch 2.0.1
- **Paper**: [Deep Residual Learning for Image Recognition](https://arxiv.org/abs/1512.03385)
- **Contact**: ml-team@example.com

## Intended Use

### Primary Uses
- General-purpose image classification
- Feature extraction for transfer learning
- Research baseline for computer vision tasks

### Primary Users
- ML researchers
- Application developers
- Computer vision practitioners

### Out-of-Scope Uses
- Medical diagnosis without additional validation
- Security/surveillance without bias auditing
```

```markdown
- High-stakes decision-making without human oversight
- Real-time critical systems (latency: ~45ms)

## Factors

### Groups
- Lighting conditions: bright, normal, dark
- Object sizes: small, medium, large
- Image quality: high-resolution, compressed
- Geographic diversity: tested on ImageNet distribution

### Instrumentation
- Tested on images from various cameras and devices
- JPEG compression levels: 75-100 quality

## Metrics

- **Accuracy**: Top-1 classification accuracy
- **Top-5 Accuracy**: Correct class in top 5 predictions
- **Inference Time**: Average milliseconds per image (on V100 GPU)
- **Model Size**: 98 MB (FP32)

## Evaluation Data

### Dataset
- **Name**: ImageNet-1K Validation Set
- **Size**: 50,000 images
- **Distribution**: 1000 classes, 50 images per class
- **Source**: http://www.image-net.org/

### Preprocessing
1. Resize shortest edge to 256 pixels
2. Center crop to 224×224
3. Normalize with ImageNet mean and std

## Training Data

### Dataset
- **Name**: ImageNet-1K Training Set
- **Size**: 1,281,167 images
- **Classes**: 1000 object categories
- **Collection**: Scraped from internet, annotated via Amazon Mechanical Turk

### Data Characteristics
- Geographic bias toward North American and European contexts
- Temporal bias (images from 2009-2011 era)
- Known class imbalances (ranging from 732 to 1300 images per class)

## Quantitative Analysis

### Overall Performance
| Metric | Value |
|--------|-------|
| Top-1 Accuracy | 76.2% |
| Top-5 Accuracy | 93.1% |
| Inference Time (GPU) | 45.2 ms |
| Inference Time (CPU) | 287 ms |

### Performance by Lighting Condition
| Condition | Top-1 Acc | Top-5 Acc |
|-----------|-----------|-----------|
| Bright | 81.2% | 95.1% |
| Normal | 76.2% | 93.1% |
| Dark | 63.1% | 84.7% |

**Finding**: Significant performance degradation in low-light conditions.

### Performance by Object Size
| Object Size | Top-1 Acc | Top-5 Acc |
|-------------|-----------|-----------|
| Large (>50% image) | 85.4% | 96.7% |
| Medium (20-50%) | 76.2% | 93.1% |
| Small (<20%) | 59.1% | 82.1% |
```

````markdown
**Finding**: Model struggles with small objects, likely due to spatial resolution loss.

### Confusion Matrix Analysis
- Most common confusions: visually similar categories (e.g., dog breeds)
- Systematic errors: texture over shape bias observed

## Ethical Considerations

### Bias and Fairness
- **Geographic bias**: Model trained predominantly on Western imagery
- **Cultural bias**: Object categories reflect Western cultural context
- **Representation gaps**: Underrepresentation of objects from Global South

### Privacy
- Training data sourced from public internet
- Potential privacy concerns with incidentally captured individuals
- No explicit consent from people in images

### Potential Harms
- **Stereotyping**: May reinforce cultural stereotypes through category definitions
- **Misclassification impacts**: Errors in sensitive applications (e.g., content moderation)
- **Dual use**: Could be misused for surveillance without appropriate safeguards

### Mitigation Strategies
- Document known biases for downstream users
- Recommend additional validation for sensitive applications
- Provide disaggregated performance metrics

## Caveats and Recommendations

### Limitations
1. **Low-light performance**: Accuracy drops significantly in dark conditions
2. **Small object detection**: Poor performance on objects <20% of image area
3. **Domain shift**: Performance degrades on images significantly different from ImageNet
4. **Temporal drift**: Training data from 2009-2011 may not reflect current visual world
5. **Class imbalance**: Some categories have better representation than others

### Recommendations
1. **Fine-tuning**: Recommended for domain-specific applications
2. **Ensemble methods**: Consider combining with other models for robust performance
3. **Human oversight**: Required for high-stakes decisions
4. **Monitoring**: Track performance metrics in production
5. **Retraining**: Consider periodic retraining on more recent data

### Edge Cases
- Adversarial examples can cause misclassification
- Unusual viewing angles may reduce accuracy
- Heavily compressed or artifacted images show degraded performance

### Usage Guidelines
```python
# Good practice
prediction, confidence = model.predict(image)
if confidence < 0.7:
    flag_for_human_review(image, prediction)

# Bad practice (avoid)
# Using without confidence threshold
# Deploying without domain-specific validation
# Using for medical diagnosis without FDA clearance
````

1.6 Model Card Authors

- Jane Doe (ML Engineer)
- John Smith (AI Ethics Researcher)
- Sarah Johnson (Data Scientist)

Last Updated: December 4, 2025
````

````
### Use-Case 2: Automated Model Card Generation

**Tool**: TensorFlow Model Card Toolkit

**Code Example**:
```python
from model_card_toolkit import ModelCardToolkit

# Initialize toolkit
mct = ModelCardToolkit()

# Create model card
model_card = mct.scaffold_assets()

# Populate model details
model_card.model_details.name = "ResNet-50 Classifier"
model_card.model_details.version.name = "1.0.0"
model_card.model_details.license = "Apache 2.0"
model_card.model_details.references = [
    "https://arxiv.org/abs/1512.03385"
]

# Add intended use
model_card.considerations.use_cases = [
    "General image classification",
    "Transfer learning feature extraction"
]
model_card.considerations.limitations = [
    "Reduced accuracy in low-light conditions",
    "Not suitable for medical diagnosis"
]

# Add quantitative analysis
from model_card_toolkit import Graphic

# Performance plot
model_card.quantitative_analysis.graphics.collection = [
    Graphic(
        name="Performance by Lighting",
        image="path/to/performance_plot.png"
    )
]

# Performance metrics
model_card.quantitative_analysis.performance_metrics = [
    {"type": "accuracy", "value": 0.762},
    {"type": "top_5_accuracy", "value": 0.931}
]

# Generate model card
mct.update_model_card(model_card)
html = mct.export_format()

print("Model card generated!")
````

1.7 Examples

1.7.1 Example 1: Complete Model Card (Markdown)

```markdown
# Model Card: Sentiment Analysis BERT

## Model Details
- **Developed by**: NLP Research Team
- **Model date**: December 2025
- **Model type**: BERT-base fine-tuned for sentiment analysis
- **Model version**: 2.1.0
```
```

```markdown
- **License**: MIT
- **Framework**: Hugging Face Transformers
- **Contact**: nlp-team@example.com

## Intended Use
### Primary Uses
- Social media sentiment analysis
- Customer review classification
- Brand monitoring

### Primary Users
- Marketing analysts
- Social media managers
- Customer service teams

### Out-of-Scope Uses
- Medical or legal document analysis
- Content moderation without human review
- Languages other than English

## Factors
### Groups
- Age demographics (Gen Z, Millennials, Gen X, Boomers)
- Platform types (Twitter, Reddit, Reviews)
- Text formality (casual, formal, slang-heavy)

## Metrics
- **F1 Score**: Macro-averaged F1
- **Accuracy**: Overall classification accuracy
- **AUC-ROC**: Area under ROC curve
- **Fairness Metrics**: Demographic parity difference

## Training Data
- **Dataset**: Mixed corpus
  - IMDB Reviews: 50,000 samples
  - Twitter Sentiment: 1.6M tweets
  - Amazon Reviews: 400,000 reviews
- **Preprocessing**: Lowercase, remove URLs, emoji handling
- **Class distribution**: Positive 45%, Negative 35%, Neutral 20%

## Quantitative Analysis

### Overall Performance
| Metric | Value |
|--------|-------|
| Accuracy | 89.2% |
| F1 Score | 0.887 |
| AUC-ROC | 0.942 |

### Performance by Platform
| Platform | Accuracy | F1 Score |
|----------|----------|----------|
| Twitter | 87.3% | 0.861 |
| Reviews | 91.8% | 0.908 |
| Reddit | 88.1% | 0.873 |

### Fairness Analysis
Demographic parity difference across age groups: 0.043 (below 0.05 threshold)

## Ethical Considerations
- **Bias**: Slight bias toward formal language
- **Sarcasm detection**: Limited capability (58% accuracy on sarcastic text)
- **Cultural context**: Trained primarily on US English

## Caveats
- Performance degrades on text >512 tokens
- Emojis and emoticons may affect accuracy
- Domain adaptation recommended for specialized industries
```

1.7.2 Example 2: JSON Model Card Schema

```
{
  "model_details": {
    "name": "Image Segmentation U-Net",
    "version": "3.0.1",
    "description": "Medical image segmentation for organ detection",
    "architecture": "U-Net with ResNet34 encoder",
    "developed_by": "Medical AI Lab",
    "date": "2025-12-04",
    "license": "CC BY-NC 4.0",
    "contact": "medical-ai@hospital.org"
  },
  "intended_use": {
    "primary_uses": [
      "Liver segmentation in CT scans",
      "Research tool for medical imaging"
    ],
    "primary_users": ["Radiologists", "Medical researchers"],
    "out_of_scope": [
      "Primary diagnostic tool without radiologist review",
      "Pediatric imaging (not validated)",
      "Real-time intraoperative use"
    ]
  },
  "factors": {
    "groups": ["Adult patients (18-80 years)", "CT scanner types", "Contrast vs non-contrast"],
    "instrumentation": ["Siemens SOMATOM", "GE Revolution CT", "Philips iCT"],
    "environment": ["Hospital setting", "Radiology department"]
  },
  "metrics": [
    {"name": "Dice Score", "description": "Overlap between prediction and ground truth"},
    {"name": "IoU", "description": "Intersection over Union"},
    {"name": "Hausdorff Distance", "description": "Boundary accuracy"}
  ],
  "training_data": {
    "datasets": [
      {
        "name": "LiTS Challenge Dataset",
        "size": 131,
        "url": "https://competitions.codalab.org/competitions/17094"
      },
      {
        "name": "Internal Hospital Dataset",
        "size": 450,
        "description": "IRB approved, de-identified CT scans"
      }
    ],
    "preprocessing": [
      "Resampling to 1mm × 1mm × 1mm",
      "Intensity windowing (40, 400 HU)",
      "Z-score normalization"
    ]
  },
  "quantitative_analysis": {
    "overall": {
      "dice_score": 0.947,
      "iou": 0.901,
      "hausdorff_distance_mm": 2.3
    },
    "by_scanner": {
      "Siemens": {"dice": 0.952, "iou": 0.909},
      "GE": {"dice": 0.945, "iou": 0.897},
      "Philips": {"dice": 0.943, "iou": 0.894}
    },
    "by_contrast": {
      "contrast_enhanced": {"dice": 0.961, "iou": 0.925},
      "non_contrast": {"dice": 0.933, "iou": 0.876}
    }
  },
  "ethical_considerations": {
    "privacy": "All training data de-identified per HIPAA",
    "bias": "Underrepresentation of pediatric and elderly patients",
```

```json
    "fairness": "Performance tested across scanner vendors",
    "potential_harms": [
      "False negatives could delay treatment",
      "False positives may cause unnecessary procedures"
    ],
    "mitigation": [
      "Always used with radiologist oversight",
      "Uncertainty quantification included in output",
      "Regular performance monitoring required"
    ]
  },
  "caveats": {
    "limitations": [
      "Not validated for pediatric patients (<18 years)",
      "Performance degrades with severe liver disease",
      "Requires contrast-enhanced CT for best results"
    ],
    "recommendations": [
      "Use as decision support tool only",
      "Requires clinical validation before deployment",
      "Monitor performance on institution-specific data",
      "Retrain with local data for optimal performance"
    ]
  }
}
```

1.8 Tools & Ecosystem

1.8.1 Model Card Creation

Tool	Description	Homepage
Model Card Toolkit	TensorFlow official toolkit	https://github.com/tensorflow/model-card-toolkit
Hugging Face Hub	Integrated model card editor	https://huggingface.co/docs/hub/model-cards
Model Card Generator	Template-based generator	https://modelcards.withgoogle.com/about
scikit-learn	Model card support (experimental)	https://scikit-learn.org/

1.8.2 Validation & Analysis

Tool	Description	Homepage
Fairlearn	Fairness assessment	https://fairlearn.org/
AI Fairness 360	IBM bias detection	https://aif360.mybluemix.net/
What-If Tool	Model probing	https://pair-code.github.io/what-if-tool/
Explainable AI	Google Cloud tool	https://cloud.google.com/explainable-ai

1.8.3 Compliance & Governance

Tool	Description	Homepage
Azure ML	Responsible AI dashboard	https://azure.microsoft.com/en-us/products/machine-learning/
Vertex AI	Google Cloud ML governance	https://cloud.google.com/vertex-ai
SageMaker	AWS model cards	https://aws.amazon.com/sagemaker/

1.9 Best Practices

1.9.1 1. Regular Updates

```
## Version History
- v1.0.0 (2025-01-15): Initial release
- v1.1.0 (2025-06-20): Added bias mitigation
- v2.0.0 (2025-12-04): Architecture update, retrained on expanded dataset
```

1.9.2 2. Include Visualizations

```python
import matplotlib.pyplot as plt

# Create performance plot
plt.figure(figsize=(10, 6))
plt.bar(factors, accuracies)
plt.title("Model Performance Across Factors")
plt.xlabel("Factor")
plt.ylabel("Accuracy")
plt.savefig("performance_by_factor.png")
```

1.9.3 3. Link to Related Documents

```
## Related Documentation
- [Technical Paper](./paper.pdf)
- [API Documentation](./api-docs.md)
- [Training Code](https://github.com/org/repo)
- [Datasheet for Dataset](./datasheet.md)
```

1.9.4 4. Use Templates

```python
# Load template
from model_card_toolkit import ModelCardToolkit

mct = ModelCardToolkit()
template = mct.scaffold_assets()

# Customize and populate
template.model_details.name = "My Model"
# ... fill in details ...

# Export
mct.export_format(template_path="custom_template.html.jinja")
```
